

Literature review on Atomic and Molecular Data for NH_3 kinetics in edge plasma for EIRENE

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I. Introduction

As reported in recent articles of research groups working within Eurofusion consortium^{1,2,3}, the use of N_2 as impurity seeds in Tokamak-devices is subject to a nitrogen balance issue: a large fraction of the nitrogen injected for the radiative cooling of the plasma is not recovered as N_2 upon regeneration using liquid helium operated cryopump. This was clearly shown and confirmed in recent global gas balance experiments at ASDEX Upgrade (AUG) and JET¹. The most probable loss channels are the implantation of nitrogen-containing ion into plasma-facing materials, nitrogen containing species co-deposition and ammonia formation³.

Since more than two decades, EIRENE model developed by ITER research groups, allowed reproducing the main trends of H_2/D_2 chemistry in Tokamak devices⁴. EIRENE makes use of optimized chemical kinetics model that describe both homogeneous gas phase chemistry and heterogeneous surface chemistry involved in the plasma-surface interaction (PSI). There was however no attempt to implement a chemical model in order to describe ammonia formation at the plasma-wall interface under tokamak edge-plasma conditions. In particular, the recent update of AMJUEL module used for kinetic data source for EIRENE global model does not include ammonia formation scheme⁵.

The objective of this research collaboration project is to build up a reduced chemical kinetic model that can be used in EIRENE in order to reach a satisfactory description, at least in terms of orders of magnitude and trends, of the homogeneous and surface chemistry of ammonia formation in tokamak edge plasmas. Considering the large number of possible elementary processes that take place when plasmas are generated in a mixture involving nitrogen and hydrogen elements (N/H plasmas), we limited our literature survey and collisional data analysis to the following species/reactions/conditions:

- Only N, N_2 , $N_2(A)$, $N_2(a')$, $N(^2D)$, NH, NH_2 , NH_3 , N^+ , N_2^+ , NH^+ , NH_2^+ , NH_3^+ , NH_4^+ species are considered. The contribution of short living radiative states in the ground state and metastable states kinetics was also evaluated and taken into account when necessary.
- Reactions linked to nitrogen ions N^+ to N^{7+} need not be included in the database, except for those reactions transforming N^+ into N.
- The plasma conditions considered are those corresponding to tokamak edge plasmas and relevant to ITER operation, i.e. a neutral pressure in the range 10^{-4} -20 Pa, a

plasma temperatures between 1 and 1000 eV, an electron density between 10^{18} - 10^{23} m^{-3} , and a surface temperature in the range 70-2000°C. The surface of the plasma-facing material is assumed to be saturated with hydrogen and nitrogen. Of course, these conditions correspond to very wide ranges of plasma parameters. This is especially the case for electron and surface temperatures. It is therefore unlikely that a single reduced model can yield an accurate quantitative description for all the targeted range of plasma conditions.

In this report, we start by presenting the most recent studies published on the modeling of collisional processes, chemical kinetics and plasma surface interaction (PSI) in non-equilibrium N/H plasmas. Then we describe the main sources of fundamental kinetic data used in the identification of the key-processes and the development of reduced models. Taking into account the wide range of the plasma conditions of interest, we analyzed the kinetics of electrons/neutral, electron-ion, ion/neutral and neutral/neutral homogeneous processes, as well as surface processes that result in ammonia formation.

II. Research strategy

The chemistry of N/H plasmas was the subject of a large interest in four main fields⁶ : Material processing, astrophysics and astrochemistry, nuclear fusion and catalytic synthesis of Ammonia from molecular nitrogen and hydrogen⁷.

Material processing: N/H low pressure non-equilibrium plasmas were extensively investigated and used in order to develop metal nitriding processes for mechanical applications, i.e., hardness improvement of cutting tools⁸, or semi-conductor industry, high- and low-k dielectrics⁹. The plasmas considered in this first field are laboratory plasmas with electron energy in the range 1-10 eV, electron densities in the range 10^8 - 10^{13} cm⁻³, working pressure in the range 1-100 Pa and a gas temperature in the range 300-1000 K. Usually high energy ions are produced in these plasmas. Both homogeneous and heterogeneous (PSI) processes were considered and addressed in this first field.

Astrophysics and astrochemistry: The chemistry in gas-phase reactions alone cannot explain the abundance of gas phase H₂, NH₃ and other complex species in interstellar clouds. A combination of gas-phase and surface chemistry on ice and dust particles was therefore invoked to explain the observed abundance for a variety of chemical species¹⁰, among which ammonia. Research in this field is nowadays very active and a theoretical effort is being devoted to improve the kinetic models¹¹⁻¹², and to design and improve advanced experimental studies of surface reactions on cosmic ice and dusts¹³. It is worthy to mention here that the data derived in the context of astrophysics and astrochemistry studies corresponds to very low temperatures, i.e., 10 K < T < 30 K, and species density, typically 10^3 - 10^5 cm⁻³ in the dense region of interstellar clouds.

Nuclear fusion: the interest for N/H plasmas in this field was first motivated by the possibility to avoid a:C-H film deposition^{14,15} and subsequent tritium retention safety issue¹⁶, when carbon was still expected as a component of ITER divertor. Preliminary laboratory plasma experiments indicated indeed that the introduction of nitrogen in C/H plasmas tends to reduce a:C-H film formation. This motivated many studies on the reduction of a:C:H film deposition by the introduction of Nitrogen using either binary or ternary mixtures of H₂, CH₄, and N₂^{17,18} or NH₃^{19,20}. These studies provided some insight in the relevant chemical processes in C/H/N plasmas under discharge conditions used to generate non-equilibrium laboratory plasmas.

Catalytic synthesis of ammonia from molecular nitrogen and hydrogen⁷: the development and optimization of the ammonia synthesis process was the main driver for a detailed investigation of surface processes involved in NH₃ catalytic formation from N₂ and H₂. The dissociation of nitrogen at the catalyst surface, i.e., dissociative chemisorption, was recognized as the limiting step for ammonia production. This motivated the use of low pressure non-equilibrium N/H plasma with the idea to provide an enhanced pre-dissociation of N₂ by either electron-impact processes or a vibrational-pumping dissociation mechanism in the gas phase.²¹

III. Summary of the most recent studies on N₂/H₂ plasma chemistry.

Although a quite significant number of articles were published on chemical models for low pressure non-equilibrium N₂/H₂ plasmas, there are only two major recent works that combine detailed experimental measurements and kinetic modeling of these plasmas.

The first one⁶ is the work published by the group of the Instituto de Estructura de la Materia, IEM-CSIC in Madrid in 2011. This group derived a mechanism of NH₃ formation in a hollow DC discharge working at pressures in the range 1-10 Pa, gas and surface temperatures around 300 K, a fairly low electron density, i.e., $n_e \approx 1\text{-}5 \times 10^{10} \text{ cm}^{-3}$, and an electron temperature in the range 1-5 eV. The mechanism they propose includes 56 homogeneous reactions, 10 heterogeneous ion-recombination processes and 15 surface chemistry processes resulting in ammonia formation. This mechanism was validated for the following plasma process conditions: low pressure (0.8–8 Pa), and a 90% H₂ + 10% N₂ initial mixture. The analysis of the measurements by mean of a kinetic model allowed identifying the main processes responsible for the observed distributions of neutrals and ions at different pressure conditions. In particular, these authors concluded that the chemistry of neutral species is dominated by the formation of a significant amount of NH₃, i.e., up to 70% of the nitrogen containing species, at the metallic wall of the reactor through the hydrogenation of atomic nitrogen and nitrogen containing radicals. They also emphasis the relative contribution of electron-impact and ion-conversion processes in the observed ion distribution.

The second study²² published in 2015 by two German groups combines experimental measurements of neutral, radical, and ion densities, as well as chemical kinetics modeling of H₂-N₂-Ar plasmas generated in capacitively coupled radiofrequency discharges at a pressure of 1.5 Pa, input power of 200W and N₂ composition ranging between 0 and 56%. Such conditions would correspond to a fairly low plasma density, i.e., below 10^{11} cm^{-3} , an electron temperature of few electron-volts and gas and surface temperatures slightly above 300 K. These groups proposed a chemical model for Ar/N₂/H₂ plasmas with additional N/H processes as compared with former studies. In particular, reactions involving N₂-metastable states are considered. The model was validated by comparison with experiments.

If these two models are the more recently published on non-equilibrium N₂/H₂ plasmas, they both often refer to a series of paper published by Gordiets et al. in the frame of collaborations with the groups of Toulouse²³⁻²⁴, Lisbon²⁵ and Bari²⁶. In particular, Gordiets et al.²⁴ proposed the most quoted surface chemistry model for NH₃ formation in N₂/H₂ plasma in 1998. This model is inspired from the work of Boudart et al.²⁷. The plasma chemistry model was validated for pressure-values ranging between 60 and 250 Pa, and the contribution of nitrogen-metastables, N₂(a'¹Σ_u⁻) and N₂(A³Σ_u⁺), in the dissociation kinetics of H₂ was clearly evidenced²⁰.

IV. Critical description of the main kinetics data sources

The analysis of the different kinetic model of N_2/H_2 plasma published in the literature shows that one may distinguish six families of elementary processes. For each of these families, two or three main data compilations, mainly based on experimental measurements, are available.

These six main families are

- electron/neutral collisions
- electron/ion dissociative recombination
- ion/neutral collisions
- metastable-states chemistry
- neutral/neutral chemistry
- surface chemistry

We give below a short description of the different sources used to build up the data compilation we propose at the end of this report.

a. Electron-neutral processes

i. Cross section data sources for electron/neutral collision

Under low pressure and weak to moderate ionization conditions, such as those encountered in tokamak edge plasmas, the plasma kinetics is usually dominated by electron-heavy species collisional processes. Kinetic data for this type of elementary processes are quite well known for atomic species and many stable molecular species such as molecular hydrogen and nitrogen. A large effort was devoted to building up fairly detailed cross section databases for these species over a wide range of electron energy. Among the available databases, some like the electron-impact cross section databases of Janev et al.²⁸ and Reiter²⁹ for molecular hydrogen, atomic hydrogen and a large number of hydrocarbon species are often quoted. They were in particular used in the model and reaction scheme derived by Amjuel⁵.

Collisional processes of electron with radical species, e.g., NH , NH_2 , or electronically excited species, e.g., $N_2(A)$, $N_2(a')$, have been however less addressed due to the difficulty of cross section measurements for these species (beam experiments). Nevertheless, theoretical calculations were performed to estimate ionization and dissociation cross section data for electronically excited states of some diatomic molecules (H_2 ³⁰ and N_2 ³¹).

Beside the initial cross-section database of Cosby et al.³² and used in the model proposed by Amjuel, consistent electron-impact cross sections sets recommended and often updated by Phelps and co-workers³³ since the 70's are of prime importance since they yield accurate prediction of electron swarm coefficients. Also of great interest are the electron-impact cross section sets published by Hayashi³⁴ in 2003 and Itikawa³⁵ more recently.

The electron-nitrogen, electron-hydrogen and electron-atom collisions cross sections sets mentioned above were recently critically reviewed, analyzed and compiled in the frame of the LXCAT project³⁶. LXCAT platform gives access to a fairly large set of cross section data for electron scattering from ground state neutral atoms and molecules (see example on NH_3

ionization process in Figure 1). These data are given over a wide energy range, i.e., from 0 to several 100's of eV and even higher for some processes. The platform gives information on total, momentum transfer, ionization, attachment, vibrational excitation (molecules) and electronic excitation cross sections. The complete set of cross sections may be used as input to a Boltzmann equation solver in order to determine the electron energy distribution function in collisional non-equilibrium plasmas. As far as electron-impact ammonia ionization is concerned, Figure 1 shows an example of output from the LXCAT platform where the cross-section data from several databases and over different energy ranges are combined for NH_3 ionization by electronic collision. This allows both comparing different sources and accessing the cross-section value over a wide range of electron energy. Note however that only one source data, i.e., Hayashi one, provides cross section data near the threshold, which is of prime importance for plasma conditions with low electron temperature values, i.e., 110 eV.

www.lxcat.net
13 Mar 2017

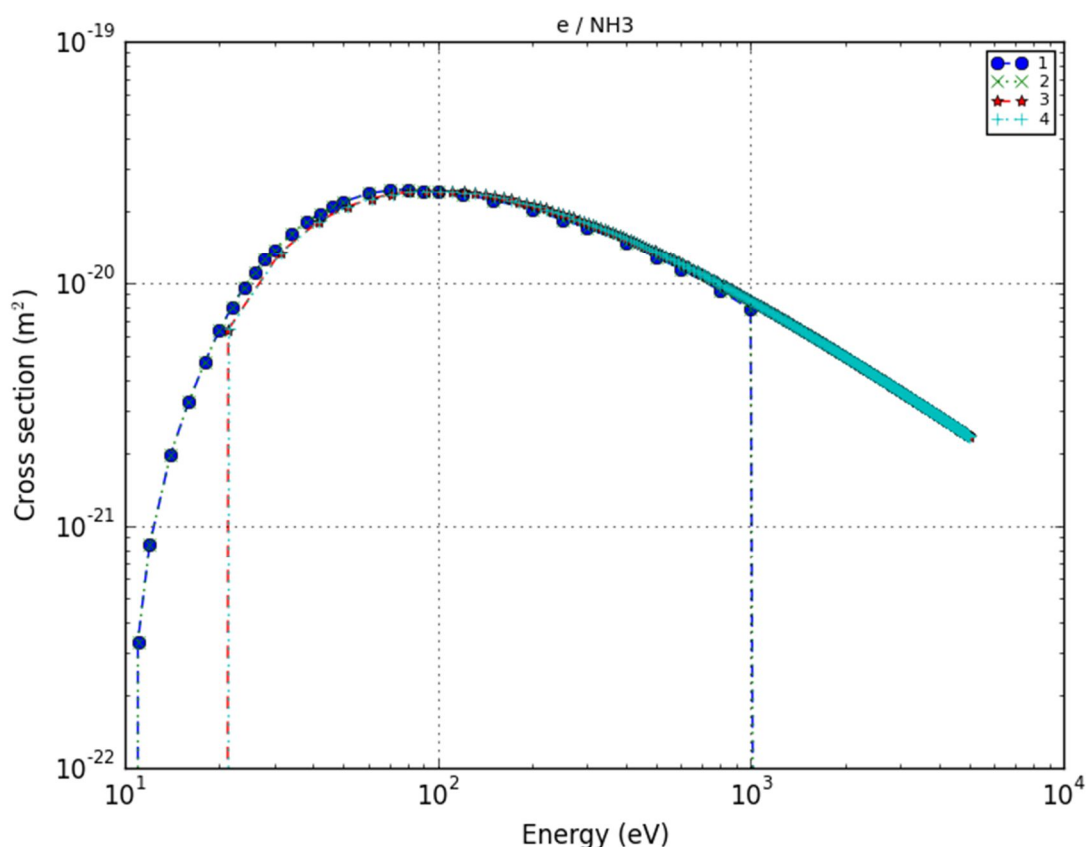


Figure 1 : Comparison between different sources cited in Lxcat Hayashi database (1), Morgan database (2), Quantemol database (3&4) for cross section in NH_3 ionization processes

ii. From electron-impact cross section data to electron-heavy species reaction rate constants

In principle, the cross-section data obtained from different database can be used to determine electron-heavy species reaction rate constants, provided the electron energy distribution function is given. Usually, the rate constants are assumed to depend only on the

electron temperature and are therefore expressed as function of this parameter using dedicated curve fits.

There are many of such curve fits expressions proposed by different authors in the literature. The most common mathematical form is the generalized Arrhenius expression, as proposed by Sode et al.²² for instance. More complex expressions involving polynomials or combination of polynomial and exponential functions are also proposed (see for example Carrasco et al.⁶). The existence of such a variety of expressions is simply due to the fact that the “exact” numerical estimation of a reaction rate constant using a realistic cross section data cannot be curve fitted by a simple mathematical expression over a very large range of electron temperature even if a Maxwellian EEDF is assumed. Therefore, simple curve fit expressions are generally only valid over a limited range of electron temperature. Derivation of such expressions to be used in Eirene simulations should be accompanied with a critical analysis of the expected accuracy over the considered range of electron temperature.

To illustrate the accuracy issue related to the use of simple rate constant expressions, figures 2 and 3 show ‘striking’ examples of the difference one may find between the estimates given by three different sources for the same rate constants for a priori two very well-known processes, i.e., H_2 and N_2 ionization processes, over a wide range of electron temperature. It appears that the rate constant expressions given by Sode et al.²² and Amjuel⁵ differ by less than one order of magnitude in the energy range 1-1000 eV for hydrogen and 5-200 eV for nitrogen. One may therefore conclude that both the Arrhenius form expressions proposed by Sode et al.²² or Amjuel⁵ et al. may be used to estimate the nitrogen ionization rate constant. The accuracy of these expressions may be however quite limited for an electron temperature above 500 eV. In the case of hydrogen, the differences between the three data sources are much more significant. We would recommend the Arrhenius form proposed by Sode et al.²² since it is in agreement with the values obtained from Amjuel⁵ for intermediate electron energy range on one hand, and it yields the steep decrease usually observed when numerically computing the collision integrals at the low energy range on the other hand (this is not the case for the expression proposed by Amjuel⁵). It remains however that the accuracy of the expression proposed by Sode et al.²² may be limited at very low and very high electron temperature.

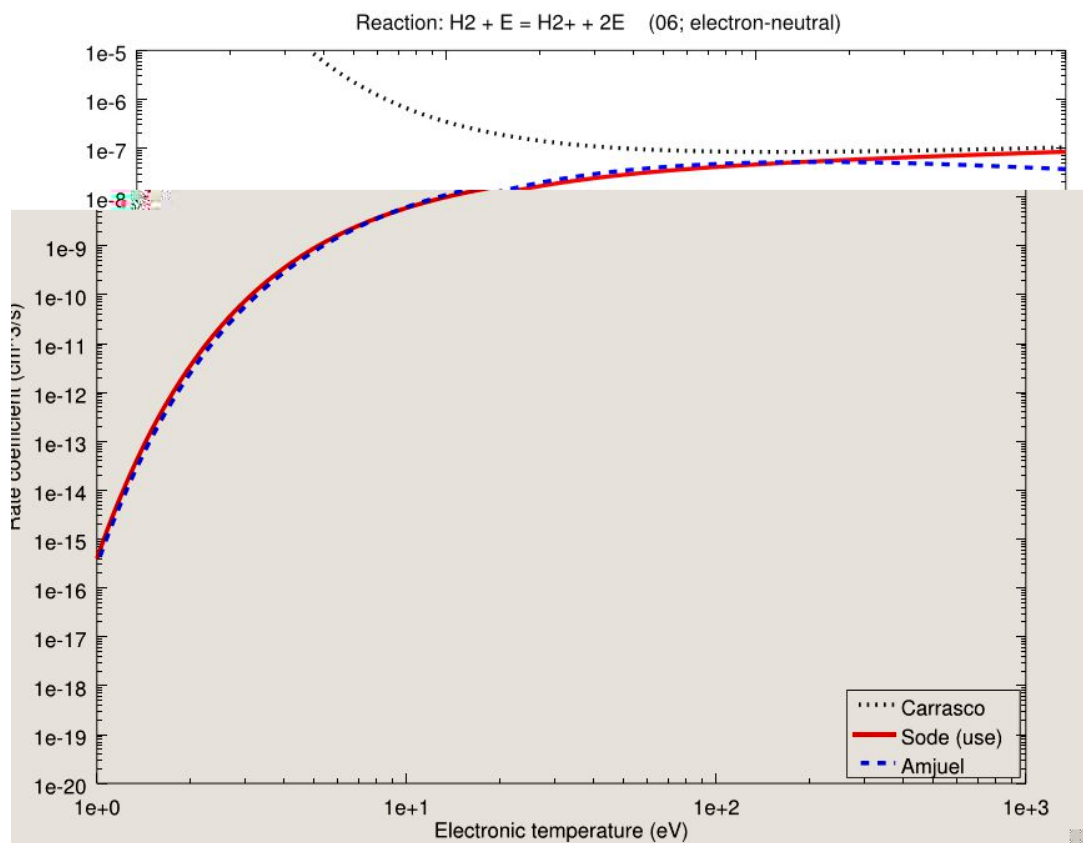


Figure 2 : Comparison of different rate coefficient data for ionization of H_2 by electron impact

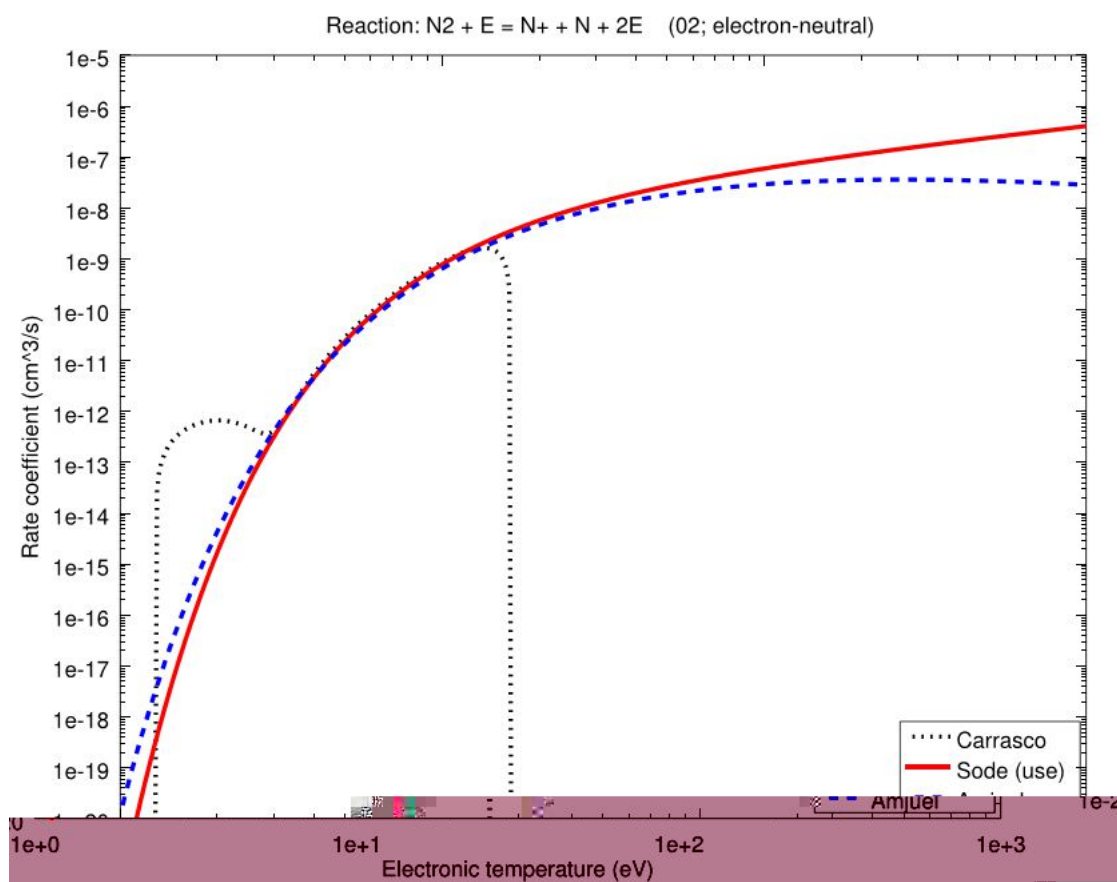


Figure 3 : Comparison of different rate coefficient data for ionization of N_2 by electron impact

iii. Rate constant data for electron-impact processes

Based on the literature survey summarized in the last sections, we identified a chemistry model that takes into account the electron-impact collisional processes reported in Table 1 and Table 2.

For the kinetic data concerning the electron impact ionization processes, we recommend the curve fit expressions proposed by Sode et al²². These are mainly obtained using the cross sections databases proposed by Itikawa³⁵.

We also suggest using generalized Arrhenius form recommended by these authors according to the expression given in equation (1), although the accuracy of these expressions may be limited at very high and very low electron energy.

$$k(T_e) = A \times T_e^n \times \exp\left(-\frac{E_a}{T_e}\right) \quad (1)$$

With A in $\text{cm}^3 \cdot \text{s}^{-1}$ and E_a and T_e in eV

This recommendation was not only motivated by the consistency of the data obtained from the Arrhenius expression with the existing data (and their validation by the authors under low pressure laboratory plasma conditions and for N_2 concentration close to those expected in SOL-PS), but also by the good agreement of the rate constant data estimated with these expressions and those obtained with the fitting parameter used in the current version of Eirene with H_2 (see Figure 2) and N_2 (see Figure 3).

Table 1 : Ionization processes collision

Reaction	Kinetic constant in $\text{cm}^3 \cdot \text{s}^{-1}$ / T_e in eV	Reference	Appendix number
$\text{N} + \text{e} \rightarrow \text{N}^+ + 2\text{e}$	$9.3 \times 10^{-9} \times T_e^{0.56} \times \exp(-16.66/T_e)$	37	1
$\text{N}_2 + \text{e} \rightarrow \text{N}^+ + \text{N} + 2\text{e}$	$2.9 \times 10^{-9} \times T_e^{0.72} \times \exp(-29.71/T_e)$	38	2
$\text{N}_2 + \text{e} \rightarrow \text{N}_2^+ + 2\text{e}$	$1.3 \times 10^{-8} \times T_e^{0.56} \times \exp(-17.07/T_e)$	38	3
$\text{H} + \text{e} \rightarrow \text{H}^+ + 2\text{e}$	$1.1 \times 10^{-8} \times T_e^{0.29} \times \exp(-15.28/T_e)$	39	4
$\text{H}_2 + \text{e} \rightarrow \text{H}^+ + \text{H} + 2\text{e}$	$9.4 \times 10^{-10} \times T_e^{0.45} \times \exp(-29.94/T_e)$	40	5
$\text{H}_2 + \text{e} \rightarrow \text{H}_2^+ + 2\text{e}$	$2.3 \times 10^{-8} \times T_e^{0.19} \times \exp(-17.87/T_e)$	40	6
$\text{NH} + \text{e} \rightarrow \text{NH}^+ + 2\text{e}$	$2.1 \times 10^{-8} \times T_e^{0.37} \times \exp(-15.49/T_e)$	41	7

Table 2 : Electron-impact dissociation

Reaction	Kinetic constant in $\text{cm}^3 \cdot \text{s}^{-1} / \text{Te in eV}$	Reference	Appendix number
$\text{H}_2 + \text{e} \rightarrow \text{H} + \text{H} + \text{e}$	$8.4 \times 10^{-8} \times \text{Te}^{-0.45} \times \exp(-11.18/\text{Te})$	40	13
$\text{N}_2 + \text{e} \rightarrow \text{N} + \text{N}(^2\text{D}) + \text{e}$	$2.4 \times 10^{-8} \times \text{Te}^{0.27} \times \exp(-15.53/\text{Te})$	38	14
$\text{NH} + \text{e} \rightarrow \text{N} + \text{H} + \text{e}$	$4.7 \times 10^{-8} \times \text{Te}^{-0.22} \times \exp(-7.69/\text{Te})$	43	15
$\text{NH}_2 + \text{e} \rightarrow \text{N} + \text{H}_2 + \text{e}$	$1.5 \times 10^{-8} \times \text{Te}^{0.38} \times \exp(-11.44/\text{Te})$		16
$\text{NH}_2 + \text{e} \rightarrow \text{NH} + \text{H} + \text{e}$	$4.5 \times 10^{-8} \times \text{Te}^{-0.22} \times \exp(-7.61/\text{Te})$		17
$\text{NH}_3 + \text{e} \rightarrow \text{NH}_2 + \text{H} + \text{e}$	$4.2 \times 10^{-8} \times \text{Te}^{-0.19} \times \exp(-7.59/\text{Te})$	44	18
$\text{NH}_3 + \text{e} \rightarrow \text{NH} + \text{H}_2 + \text{e}$	$4.1 \times 10^{-8} \times \text{Te}^{-0.26} \times \exp(-4.81/\text{Te})$		19
$\text{NH}_3 + \text{e} \rightarrow \text{NH} + 2\text{H} + \text{e}$	$1.3 \times 10^{-8} \times \text{Te}^{0.38} \times \exp(-11.06/\text{Te})$		ZE

b. Electron-ion recombination processes

The procedure adopted for these processes is similar to the one used in the case of electron-neutral processes. It is however worthy to mention here that unlike the activated ionization and dissociation processes, ion-recombination processes usually show monotonously decreasing cross sections as function of the electron energy.⁴⁵ They are therefore not activated, $E_a \approx 0$ eV, and their rate constants decrease with electron temperature. The n exponent in the pre-exponential factor is generally close to -0.5 for these processes⁴⁶.

The rate constant data for these processes are readily found in up-to-date databases, such as the University of Manchester Institute of Science and Technology (UMIST) database, and the recent articles published low pressure N/H plasmas modeling and mentioned above. Most of the data published for these processes are determined experimentally at ambient temperature. Some of them were confirmed by recent theoretical calculations.⁴⁵ The process and rate constants identified in these sources are listed in Table 3. The corresponding references are given in the “reference” column of this table.

Table 3 : Dissociative recombination processes

Reaction	Kinetic constant in $\text{cm}^3 \cdot \text{s}^{-1}$	Reference	Appendix number
$\text{H}_2^+ + \text{e} \rightarrow \text{H} + \text{H}$	$1.60 \times 10^{-8} \times (\text{Te}/0.026)^{-0.43}$	47	20
$\text{H}_3^+ + \text{e} \rightarrow 3\text{H}$	$4.36 \times 10^{-8} \times (\text{Te}/0.026)^{-0.52}$	48	21
$\text{H}_3^+ + \text{e} \rightarrow \text{H}_2 + \text{H}$	$2.34 \times 10^{-8} \times (\text{Te}/0.026)^{-0.52}$	48	22
$\text{N}_2^+ + \text{e} \rightarrow \text{N}(^2\text{D}) + \text{N}$	$1.70 \times 10^{-7} \times (\text{Te}/0.026)^{-0.3}$	49	23
$\text{NH}^+ + \text{e} \rightarrow \text{N} + \text{H}$	$4.30 \times 10^{-8} \times (\text{Te}/0.026)^{-0.5}$	47	24
$\text{NH}_2^+ + \text{e} \rightarrow \text{NH} + \text{H}$	$1.02 \times 10^{-7} \times (\text{Te}/0.026)^{-0.4}$	50	25
$\text{NH}_2^+ + \text{e} \rightarrow \text{N} + 2\text{H}$	$1.98 \times 10^{-8} \times (\text{Te}/0.026)^{-0.4}$	50	26
$\text{HN}_2^+ + \text{e} \rightarrow \text{N}_2 + \text{H}$	$1.3 \times 10^{-8} \times (\text{Te}/0.026)^{-0.84}$	51	ZF
$\text{NH}_3^+ + \text{e} \rightarrow \text{NH} + 2\text{H}$	$1.55 \times 10^{-7} \times (\text{Te}/0.026)^{-0.5}$	47	27
$\text{NH}_3^+ + \text{e} \rightarrow \text{NH}_2 + \text{H}$	$1.55 \times 10^{-7} \times (\text{Te}/0.026)^{-0.5}$	47	28
$\text{NH}_4^+ + \text{e} \rightarrow \text{NH}_3 + \text{H}$	$8.49 \times 10^{-7} \times (\text{Te}/0.026)^{-0.6}$	50	29
$\text{NH}_4^+ + \text{e} \rightarrow \text{NH}_2 + 2\text{H}$	$3.77 \times 10^{-8} \times (\text{Te}/0.026)^{-0.6}$	50	30
$\text{N}_2\text{H}^+ + \text{e} \rightarrow \text{N}_2 + \text{H}$	$5.13 \times 10^{-8} \times (\text{Te}/0.026)^{-0.72}$	52	31

c. Ion-neutral processes

The review published by Anicich et al. in 1993⁵³ is probably the major data source used by most of the authors during the last two decades as far as ion/neutral collisions are concerned. This review was updated by the same authors in 2003⁵⁴. It provides a large number of data with a thorough critical analysis based on comparison between at least two or three data sources that generally rely on experimental measurements for each process.

It is however worthy to mention that most of the data recommended in Anicich review are mainly centered on works of W.T. Huntress and co-workers in the 70's⁵⁵. To expand the number of possible data sources, we therefore compare the data recommended by Anicich with those suggested in the database of UMIST⁵⁶. This has been developed for more than two decades. It is mainly devoted to modeling phenomena in interstellar systems. The last update of 2012 contains 6173 gas-phase reactions, involving 467 species. In this database, the temperature dependence of the rate constant is not considered and only constant values are given for most of the ion-neutral reaction rate constants. Note however that the rate constants are assumed to scale as $T_g^{-0.5}$ for some processes. This scaling is classically used to account for the interaction between the permanent dipole of the neutral species and the electrical charge of the ion. This effect is significant at low temperature, i.e., below 1000 K, and when the permanent dipole of the neutral species exceeds 0.9 Debye. It may result in a large increase of the kinetic constant at low temperature due to a 'dipole locking' phenomenon⁵⁷.

The Arrhenius form expression giving the rate constants as function of the gas temperature, T_g , are compiled in Table 4. When several data sources are available for a given reaction, their number is mentioned in the column "*Reference*". The ion/neutral processes and reaction rate data selected in this work come from Anicich et al.⁵³ and have been recently confirmed by studies combining experimental investigation and kinetic modeling of non-equilibrium laboratory plasmas.

Table 4 : Processes and rate coefficients for ion/neutral collisional processes

Reaction	Kinetic constant in $\text{cm}^3.\text{s}^{-1}$ with T_g in eV	Reference	Appendix number
$\text{H}^+ + \text{NH}_3 \rightarrow \text{NH}_3^+ + \text{H}$	$3.7 \times 10^{-09} \times (T_g/0.026)^{-0.5}$	58	32
$\text{H}^+ + \text{NH}_2 \rightarrow \text{NH}_2^+ + \text{H}$	$2.9 \times 10^{-09} \times (T_g/0.026)^{-0.5}$	59	ZL
$\text{H}^+ + \text{NH} \rightarrow \text{NH}^+ + \text{H}$	$2.1 \times 10^{-09} \times (T_g/0.026)^{-0.5}$		ZM
$\text{H}_2^+ + \text{H} \rightarrow \text{H}_2 + \text{H}^+$	6.4×10^{-10}	60	33
$\text{H}_2^+ + \text{H}_2 \rightarrow \text{H}_3^+ + \text{H}$	2.00×10^{-09}	53 evaluated from 7 ref.	34
$\text{H}_2^+ + \text{NH}_3 \rightarrow \text{NH}_3^+ + \text{H}_2$	5.70×10^{-09}	61	35
$\text{H}_2^+ + \text{N}_2 \rightarrow \text{N}_2\text{H}^+ + \text{H}$	2.00×10^{-09}	53 evaluated from 3 ref.	36
$\text{H}_2^+ + \text{N} \rightarrow \text{N}^+ + \text{H}_2$	5.00×10^{-10}	62	ZN
$\text{H}_2^+ + \text{N} \rightarrow \text{NH}^+ + \text{H}$	1.9×10^{-09}	59	ZO

$\text{H}_2^+ + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{H}$	5.00×10^{-11}	62	ZP
$\text{H}_2^+ + \text{NH}_2 \rightarrow \text{NH}_3^+ + \text{H}$	5.00×10^{-11}	62	ZQ
$\text{H}_2^+ + \text{NH}_2 \rightarrow \text{NH}_2^+ + \text{H}_2$	$2.1 \times 10^{-09} \times (\text{Tg}/0.026)^{-0.5}$	59	ZR
$\text{H}_2^+ + \text{NH} \rightarrow \text{NH}_2^+ + \text{H}$	$7.6 \times 10^{-10} \times (\text{Tg}/0.026)^{-0.5}$	59	ZS
$\text{H}_2^+ + \text{NH} \rightarrow \text{NH}^+ + \text{H}_2$	$7.6 \times 10^{-10} \times (\text{Tg}/0.026)^{-0.5}$	59	ZT
$\text{H}_3^+ + \text{N} \rightarrow \text{NH}^+ + \text{H}_2$	2.60×10^{-10}	53 evaluated from one ref of huntress	37
$\text{H}_3^+ + \text{N} \rightarrow \text{NH}_2^+ + \text{H}$	3.90×10^{-10}	53 evaluated from one ref of huntress	38
$\text{H}_3^+ + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{H}_2$	$4.40 \times 10^{-09} \times (\text{Tg}/0.026)^{-0.5}$	53 evaluated from 5 ref	39
$\text{H}_3^+ + \text{NH}_2 \rightarrow \text{NH}_3^+ + \text{H}_2$	$1.8 \times 10^{-09} \times (\text{Tg}/0.026)^{-0.5}$	59	ZU
$\text{H}_3^+ + \text{NH} \rightarrow \text{NH}_2^+ + \text{H}_2$	$1.3 \times 10^{-09} \times (\text{Tg}/0.026)^{-0.5}$	59	ZV
$\text{H}_3^+ + \text{N}_2 \rightarrow \text{N}_2\text{H}^+ + \text{H}_2$	1.90×10^{-09}	53 evaluated from 11 ref	40
$\text{N}^+ + \text{H}_2 \rightarrow \text{NH}^+ + \text{H}$	5.00×10^{-10}	53 evaluated from 11 ref	41
$\text{N}^+ + \text{NH}_3 \rightarrow \text{NH}_2^+ + \text{NH}$	4.70×10^{-10}	53 evaluated from 3 ref	42
$\text{N}^+ + \text{NH}_3 \rightarrow \text{NH}_3^+ + \text{N}$	1.70×10^{-09}	53 evaluated from 3 ref	43
$\text{N}^+ + \text{NH}_3 \rightarrow \text{N}_2\text{H}^+ + \text{H}_2$	2.10×10^{-10}	53 evaluated from 3 ref	44
$\text{N}^+ + \text{NH} \rightarrow \text{N}_2^+ + \text{H}$	$3.7 \times 10^{-10} \times (\text{Tg}/0.026)^{-0.5}$	59	ZW
$\text{N}^+ + \text{NH} \rightarrow \text{NH}^+ + \text{N}$	$3.7 \times 10^{-10} \times (\text{Tg}/0.026)^{-0.5}$	59	ZX
$\text{N}^+ + \text{N}_2 \rightarrow \text{N}_2^+ + \text{N}$	2.00×10^{-11}	63	ZY
$\text{N}^+ + \text{H} \rightarrow \text{H}^+ + \text{N}$	2.00×10^{-10}	62	ZZ
$\text{N}^+ + \text{NH}_2 \rightarrow \text{NH}_2^+ + \text{N}$	1.0×10^{-09}	59	AB
$\text{NH}^+ + \text{H}_2 \rightarrow \text{H}_3^+ + \text{N}$	1.85×10^{-10}	53 evaluated from 3 ref	45
$\text{NH}^+ + \text{H}_2 \rightarrow \text{NH}_2^+ + \text{H}$	1.00×10^{-09}	53 evaluated from 3 ref	46

$\text{NH}^+ + \text{NH}_3 \rightarrow \text{NH}_3^+ + \text{NH}$	1.80×10^{-09}	53 evaluated from 3 ref	47
$\text{NH}^+ + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{N}$	6.00×10^{-10}	53 evaluated from 3 ref	48
$\text{NH}^+ + \text{NH}_2 \rightarrow \text{NH}_3^+ + \text{N}$	$1.5 \times 10^{-09} \times (\text{Tg}/0.026)^{-0.5}$	59	AC
$\text{NH}^+ + \text{NH}_2 \rightarrow \text{NH}_2^+ + \text{NH}$	1.80×10^{-09}	62	AD
$\text{NH}^+ + \text{NH} \rightarrow \text{NH}_2^+ + \text{N}$	1.0×10^{-09}	59	AE
$\text{NH}^+ + \text{N}_2 \rightarrow \text{N}_2\text{H}^+ + \text{N}$	6.50×10^{-10}	64	49
$\text{NH}^+ + \text{N} \rightarrow \text{N}_2^+ + \text{H}$	1.3×10^{-09}	59	AG
$\text{NH}_2^+ + \text{H}_2 \rightarrow \text{NH}_3^+ + \text{H}$	2.00×10^{-10}	53 evaluated from 3 ref	50
$\text{NH}_2^+ + \text{NH}_3 \rightarrow \text{NH}_3^+ + \text{NH}_2$	1.20×10^{-09}	53 evaluated from 3 ref	51
$\text{NH}_2^+ + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{NH}$	1.20×10^{-09}	53 evaluated from 3 ref	52
$\text{NH} + \text{NH}_2^+ \rightarrow \text{NH}_3^+ + \text{N}$	$7.3 \times 10^{-10} \times (\text{Tg}/0.026)^{-0.5}$	59	AH
$\text{NH}_2^+ + \text{N} \rightarrow \text{HN}_2^+ + \text{H}$	9.1×10^{-11}	59	AI
$\text{NH}_3^+ + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{NH}_2$	2.10×10^{-09}	53 evaluated from 3 ref	53
$\text{NH}_3^+ + \text{H}_2 \rightarrow \text{NH}_4^+ + \text{H}$	4.40×10^{-13}	48 evaluated from 11 ref	AJ
$\text{NH}_3^+ + \text{NH} \rightarrow \text{NH}_4^+ + \text{N}$	$7.1 \times 10^{-10} \times (\text{Tg}/0.026)^{-0.5}$	59	AK
$\text{N}_2^+ + \text{H}_2 \rightarrow \text{NH}_2^+ + \text{H}$	2.00×10^{-09}	53 evaluated from 3 ref	54
$\text{N}_2^+ + \text{NH}_3 \rightarrow \text{N}_2 + \text{NH}_3^+$	2.00×10^{-09}	53 evaluated from 3 ref	55
$\text{N}_2^+ + \text{N} \rightarrow \text{N}^+ + \text{N}_2$	1.00×10^{-11}	53 evaluated from 3 ref	AL
$\text{N}_2\text{H}^+ + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{N}_2$	2.30×10^{-09}	53 evaluated from 3 ref	56
$\text{N}_2\text{H}^+ + \text{H}_2 \rightarrow \text{H}_3^+ + \text{N}_2$	5.10×10^{-18}	53 evaluated from 3 ref	AM

d. Metastable-states chemistry

The long-term work carried out by Goudier in collaboration with the groups of Toulouse, Lise, and others on the kinetic modelling of N_2/H non-equilibrium microwave plasmas is certainly one of the most important studies as far as plasma-specific processes, involving vibrationally and electronically excited states, are concerned. The chemistry models developed by these authors were at first directly validated under laboratory plasma conditions, i.e., $n_e=10^8-10^9\text{ cm}^{-3}$, $T_e=1-5\text{ eV}$, $T_g=300-1000\text{ K}$, $10-200\text{ W}$, $P_{\text{inp}}=1\text{ W}$. Goudier et al. (1995) have then extended their modelling to the

Table 6 : Processes and rate coefficients for quenching of N₂ metastable states by neutrals

Reactions	Kinetic constant in cm ³ .s ⁻¹ / Tg in eV	Ref.
N ₂ (A)+N(⁴ S) → N ₂ (X)+N(⁴ S)	2.00 × 10 ⁻¹²	67
N ₂ (A)+N(⁴ S) → N ₂ (X)+N(² P)	1.80 × 10 ⁻⁰⁹ × (Tg/0.026) ^{-0.67}	68 69 70
N ₂ (A)+N ₂ (X) → 2N ₂ (X)	3.01 × 10 ⁻¹⁰	71
N ₂ (A)+NH ₃ → N ₂ (X)+H+NH ₂	8.51 × 10 ⁻¹¹	72
N ₂ (A)+H → N ₂ (X)+H	2.09 × 10 ⁻¹⁰	73
N ₂ (A)+H ₂ → N ₂ (X) + 2H	2.00 × 10 ⁻¹⁰ × exp(-0.3016/Tg)	7475
N ₂ (A)+N ₂ (A) → N ₂ (X)+N ₂ (B)	3.01 × 10 ⁻¹⁰	76-77
N ₂ (A)+N ₂ (A) → N ₂ (X)+N ₂ (C)	1.50 × 10 ⁻¹⁰	78
N ₂ (A)+N ₂ (X, v > 6) → N ₂ (B)+N ₂ (X, v- 6)	3.01 × 10 ⁻¹¹	79
N ₂ (a')+N ₂ (X) → N ₂ (B)+N ₂ (X)	1.90 × 10 ⁻¹³	80- 81
N ₂ (a') + H → N ₂ (X)+H	1.50 × 10 ⁻¹⁰	23
N ₂ (a') + H ₂ → N ₂ (X) + 2H	2.61 × 10 ⁻¹¹	81
N ₂ (a') + M → N ₂ (a)+M	1.80 × 10 ⁻¹¹ × exp(-0.1464/Tg)	82
N ₂ (a)+M → N ₂ (a') +M	9.00 × 10 ⁻¹²	82
N ₂ (v > 16) + N ₂ (v >= 16) → N ₂ (a')+N ₂ (X) ^a	2.00 × 10 ⁻¹⁴ × exp(-0.1465/Tg)	23

Table 7 : Processes and rate coefficients for quenching of N metastable states by neutrals.

Reactions	Kinetic constant in cm ³ .s ⁻¹ / Tg in eV	Ref.
N(² D)+N ₂ (X) → N ₂ (X)+N(⁴ S) ^b	6.00 × 10 ⁻⁰⁹	46, 83
N(² D)+H ₂ → NH + H	2.29 × 10 ⁻¹²	84
N(² D)+NH ₃ → NH+NH ₂	1.10 × 10 ⁻¹⁰	83
N(² P)+N(⁴ S) → N(² D)+N(⁴ S)	1.79 × 10 ⁻¹²	46
N(² P)+N ₂ (X) → N(⁴ S)+N ₂ (X)	1.99 × 10 ⁻¹⁸	85
N(² P)+H ₂ → NH + H	2.51 × 10 ⁻¹⁴	86
N+N+N ₂ → N ₂ (A)+N ₂	1.02 × 10 ⁻⁰⁹	87,88 89
N+N+N → N ₂ (A)+N	1.02 × 10 ⁻¹⁴	

^a N₂(X) is the ground state of N₂

^b N(⁴S) is the ground state of N

e. Neutral/neutral chemistry

For the reactions between ground state neutral species produced in N/H plasmas, a global comprehensive mechanism such as those proposed by Capitelli et al.²⁵ in their book published in 2000 can include up to 82 reactions. However, since our objective is to build up a simplified mechanism that mainly describes low pressure plasma conditions, we suggest to use the mechanism proposed by Gordiets et al.²³ This mechanism contains 17 processes and was validated and adopted by many research groups working on non-equilibrium low pressure plasmas.

Note however that we have checked and confirmed this mechanism and the related collisional data by comparison with data recommended in two other databases: the NIST database available online⁹⁰ and the database of Baulch et al.⁹¹, the last update of which date back to 2005

Finally, to be consistent with our choice for metastable states reactions, we selected the processes given in Gordiets short kinetic scheme except for the three body reactions that were not considered due to the very low pressure conditions. The neutral/neutral processes and corresponding rate constant data that were selected are listed in Table 8.

Table 8 : Processes and rate coefficients for neutral/neutral reactions

Reaction	Kinetic constant in $\text{cm}^3 \cdot \text{s}^{-1} / \text{Tg}$ in eV	Reference	Appendix number
$\text{H}_2 + \text{N} \rightarrow \text{NH} + \text{H}$	$4.65 \times 10^{-11} \times \exp(-1.4304/\text{Tg})$	92	AQ
$\text{N} + \text{NH} \rightarrow \text{N}_2 + \text{H}$	5.0×10^{-11}	93	AW
$\text{H} + \text{NH} \rightarrow \text{N} + \text{H}_2$	$5.4 \times 10^{-11} \times \exp(-0.0142/\text{Tg})$	94	AV
$\text{NH} + \text{NH} \rightarrow \text{N}_2 + \text{H}_2$	$5 \times 10^{-14} \times (\text{Tg}/0.026)^{+1.0}$	94	
$\text{NH} + \text{NH} \rightarrow \text{NH}_2 + \text{N}$	$1.7 \times 10^{-12} \times (\text{Tg}/0.026)^{+1.5}$	97	AY
$\text{NH} + \text{NH} \rightarrow \text{N}_2 + 2\text{H}$	8.5×10^{-11}	95	AX
$\text{H} + \text{NH}_2 \rightarrow \text{NH} + \text{H}_2$	$6.6 \times 10^{-11} \times \exp(-0.1586/\text{Tg})$	94	AT
$\text{N} + \text{NH}_2 \rightarrow \text{N}_2 + \text{H}$	1.2×10^{-10}	93	
$\text{N} + \text{NH}_2 \rightarrow \text{N}_2 + \text{H}_2$	1.2×10^{-10}	93	
$\text{NH} + \text{NH}_2 \rightarrow \text{NH}_3 + \text{N}$	1.66×10^{-12}	97	
$\text{H}_2 + \text{NH}_2 \rightarrow \text{NH}_3 + \text{H}$	$5.4 \times 10^{-11} \times \exp(-0.5594/\text{Tg})$	96	AR
$\text{H} + \text{NH}_3 \rightarrow \text{NH}_2 + \text{H}_2$	$8.4 \times 10^{-14} \times (\text{Tg}/0.026)^{+4.1} \times \exp(-0.4102/\text{Tg})$	97	AU
$\text{H}_2 + \text{NH} \rightarrow \text{NH}_2 + \text{H}$	$5.96 \times 10^{-11} \times \exp(-0.6706/\text{Tg})$	98	AS

f. Surface reactions

There are basically three groups of reactions as far as surface chemistry under N_2/H_2 plasma conditions is concerned. These are: (i) Wall recombination of positive ions, (ii) metastable de-excitation at the wall and (iii) neutral chemistry and ammonia formation

(i) Wall recombination of positive ions

As suggested by many authors, we choose for this type of processes to follow the classical theory of wall-recombination of charged species as described in the reference book of M. A. Lieberman and A. J. Lichtenberg⁹⁹. Ions reaching the wall are supposed to recombine with electron of the wall with a probability of 1. Usually the recombination is dissociative and the resulting neutral fragments are assumed to be immediately remitted in the gas with a probability of 1. The kinetic scheme for this type of process is listed in Table 9

Table 9 : Kinetic scheme for ion neutralization on metallic wall

Wall neutralisation processes	
$H^+ + \text{wall} \rightarrow H$	$NH^+ + \text{wall} \rightarrow N + H$
$H_2^+ + \text{wall} \rightarrow H + H$	$NH_2^+ + \text{wall} \rightarrow NH + H$
$H_3^+ + \text{wall} \rightarrow H_2 + H$	$NH_3^+ + \text{wall} \rightarrow NH_2 + H$
$N^+ + \text{wall} \rightarrow N$	$NH_4^+ + \text{wall} \rightarrow NH_3 + H$
$N_2^+ + \text{wall} \rightarrow N + N$	$N_2H^+ + \text{wall} \rightarrow N_2 + H^+$

(ii) Metastable Wall-relaxation

The wall-de-excitation of metastable states may be more or less effective depending on the ability of the molecule-metal system to relax the fairly high amount of energy produced during the de-excitation process. Surprisingly, a short literature survey shows that the wall de-excitation probability of metastables is pressure dependent and may vary from 10^{-3} at atmospheric pressure to 1 at low pressure. Since we are mainly interested in very low pressure conditions, we recommend a de-excitation coefficient of 1 for both $N_2(A)$ metastables. We would also recommend to assume dissociative de-excitation for $N_2(A)$.

It is more difficult to state on the value of the de-excitation probability of $N(^2D)$ at the wall. As a matter of fact, the metastable-metal system does not show a simple metastable energy relaxation route during metastable-wall interaction in this case. The de-excitation probability of $N(^2D)$ may be therefore smaller than 1. Nevertheless, in the absence of any even qualitative information on the wall-de-excitation of this metastable, we would suggest to assume a probability of 1 for its de-excitation.

(iii) Surface chemistry of neutral species and NH_3 formation mechanism

The chemistry of ammonia production by heterogeneous catalytic reaction between N_2 and H_2 was thoroughly investigated during the 70's by surface science, catalysis and industrial chemistry communities (ammonia synthesis was a key-chemical industrial process at that time)¹⁰⁰. Among the many groups that addressed this process, the long term thorough and very high quality studies of Ertl¹⁰⁰ et al. and Somorjai et al.¹⁰¹ enabled shedding the light on the dynamic and kinetics of the gas-surface interaction processes involved in the formation of ammonia on a variety of model mono-crystalline iron surfaces. It is now widely accepted

that ammonia formation proceeds through two classical mechanisms: the Langmuir Hinshelwood mechanism, where ammonia formation results from the reaction between two adsorbed species and the Eley-Rideal mechanism where ammonia formation proceeds through a successive reaction between several gas phase H-atom and nitrogen containing adsorbed species.

Beside the fundamental studies mentioned above, a thorough investigation of the catalytic efficiency of different materials was carried out by the group of Venugopalan from Western Illinois University¹⁰². This group clearly showed that the catalytic efficiency of ammonia formation varies with the nature of the wall. The group reported the following trends of catalytic efficiency for different materials: Pt > Ag > Fe > Cu > Al > Zn.

Nevertheless, to our best knowledge, there has been no attempt in the literature to transpose the pioneering work of Ertl et al.¹⁰⁰ and Somorjai et al.¹⁰¹ on the detailed mechanism of ammonia formation to non-equilibrium plasma conditions. Most of the studies on ammonia formation in plasmas were conducted under post discharge conditions, in the absence of any ion bombardment that may significantly affect surface processes, which makes the conclusion drawn on the basis of post-discharge experiments not completely transferable to plasma conditions.

In any case among the mechanisms proposed in the literature, we focused on three main groups of sources:

- The Langmuir Hinshelwood mechanism proposed by Ertl and co-worker¹⁰⁰
- The detailed surface chemistry of Gordiets and co-workers²⁴
- The simplified surface mechanism suggested by Carrasco and al.⁶

The first group of articles corresponds to the previously mentioned long-term work of Gerhard Ertl (The 2007 Nobel Prize in Chemistry)¹⁰³ on N₂, H₂ and NH₃ surface chemistry¹⁰⁰. This work was summarized in a book published in 1991, where the surface chemistry of H₂ and N₂ on iron surface was thoroughly analyzed. The authors also showed how the ammonia formation mechanism and related kinetic data change with the pressure, the wall temperature and the crystallographic orientation of the surface. The major conclusions of this work are:

- N₂ may weakly adsorb in two adsorbed states with adsorption energies of 21 kJ.mol⁻¹ and 31 kJ.mol⁻¹ and sticking coefficients of 0.5 and 0.01, respectively.
- Dissociative chemisorption probability of molecular nitrogen on zero coverage monocrystalline surface range between 10⁻⁷ and 5x10⁻⁶ depending on the crystallographic orientation of the surface.
- Due to this low chemisorption probability, the Nitrogen adsorption is the limiting step as far ammonia catalytic production is concerned.
- Dissociative Chemisorption of N₂ leads to a strongly bonded N-atom, the thermal desorption of which only occurs above 700 K. Also, the mobility of N-atom becomes appreciable only above 500 K.

- At zero coverage, the sticking coefficient for hydrogen-molecule dissociative adsorption is equal to 0.1, which is more than five orders of magnitude larger than in the case of nitrogen.
- The variation of hydrogen sticking coefficient as a function of the surface coverage Θ , follows a Langmuir-type two-site adsorption process: $S = S_0 \times (1 - \Theta)^2$
- Surface mobility and desorption kinetics of H-atom are also much larger than their counterparts for nitrogen. Basically, desorption is completed at 500 K. Above this temperature, the surface coverage by H ad-atom is governed by adsorption-desorption equilibrium.
- Ammonia may form following a L-H mechanism by three successive hydrogenation of an almost immobile N ad-atom by 3 highly mobile H ad-atom even at fairly low temperature, i.e., below 500 K, provided nitrogen ad-atom can be formed.
- Adsorption energy of NH_3 on Fe is around 70 kJ.mol^{-1}
- Ammonia may decompose to form adsorbed NH at the surface even at fairly moderate temperature: 350-400 K.

Note that the conclusions listed above would indicate that under plasma environment, one may expect E-R production of ammonia through hydrogenation of adsorbed N, NH and NH_2 by gas phase H-atom.

On the basis of the above findings, Ertl and coworkers derived an ammonia production model based on the following assumptions: (1) The sticking coefficient for hydrogen-molecule dissociative adsorption is equal to 0.1, which is more than five orders of magnitude larger than in the case of nitrogen. (2) The variation of hydrogen sticking coefficient as a function of the surface coverage Θ , follows a Langmuir-type two-site adsorption process: $S = S_0 \times (1 - \Theta)^2$ (3) Surface mobility and desorption kinetics of H-atom are also much larger than their counterparts for nitrogen. Basically, desorption is completed at 500 K. Above this temperature, the surface coverage by H ad-atom is governed by adsorption-desorption equilibrium. (4) Ammonia may form following a L-H mechanism by three successive hydrogenation of an almost immobile N ad-atom by 3 highly mobile H ad-atom even at fairly low temperature, i.e., below 500 K, provided nitrogen ad-atom can be formed. (5) Adsorption energy of NH_3 on Fe is around 70 kJ.mol^{-1} (6) Ammonia may decompose to form adsorbed NH at the surface even at fairly moderate temperature: 350-400 K.

- deso

The kinetic values suggested by Ertl for the different steps of the Langmuir Hinshelwood mechanism are available in Table 10. The kinetics of adsorption processes for gas phase species is described in terms of an initial zero-coverage sticking coefficient, γ , the variation of which as a function of the wall-temperature is also given by an Arrhenius form expression:

$$\gamma = \gamma_0 \times \exp\left(-\frac{E_a}{T_w}\right)$$

Where γ_0 and E_a (in K) are the pre-exponential factor and the activation energy of the adsorption process, respectively. T_w is the wall temperature.

The kinetics of reactions between adsorbed species is described in terms of reaction frequencies in s^{-1} .

One of the difficulties that arise when using Ertl model comes from the lack of data for hydrogenation processes that involve two ad-atoms and lead to the formation of NH_3 . This difficulty may be circumvented by considering that these processes are very fast and that ammonia formation is essentially limited by adsorption processes. This seems to be the option implicitly proposed by Ertl et al.

An alternative option, is to use the approach of Kim and Boudart ²⁷, who suggest that the reaction frequency between two species adsorbed on a metallic surface depends on :

- the diffusion frequency of an atom on the metallic surface ν_D . This is typically around $10^{13}s^{-1}$, a value that corresponds to the surface phonon frequency according to the theory of surface diffusion proposed by Barth¹⁰⁴
- the activation temperature, E_d/R , for the surface diffusion of H-atom is approximately 2300 K.
- the activation temperature, E_r/R , of the reaction between the two adsorbed species

The reaction frequency between two adsorbed species is given by:

$$\nu = \frac{\nu_D}{4} \times \exp\left(-\frac{E_D + E_r}{T_w}\right)$$

The activation temperatures for the diffusion process of H-atom and N-atom on iron are approximately 2300 K and 10400 K, respectively. The activation energy of NH_{ads} and NH_{2-ads} by hydrogen ad-atoms are obtained from the energy diagram reported by Ertl and depicted in figure 4. The NH_{ads} and NH_{2-ads} hydrogenation processes shows activation energy of 0.3 eV and 0.2 eV, respectively.

If the new species formed by diffusion on the surface is desorbed, a third factor is involved in the computation of the desorption kinetic rate: the desorption frequency of an atom from the metallic surface ν_d . This factor is typically around $10^{15}s^{-1}$ according to Kim and Boudart ²⁷

Table 10: Sticking coefficient and surface adsorption energy for surface processes occurring in the Langmuir-Hinshelwood mechanism proposed by Ertl¹⁰⁰ as a function of wall temperature (T_w)

Surface Processes	Surface type	Initial sticking coefficient γ or reaction frequency ν (s^{-1}) with T_w in eV	Adsorption energy E_{ads} (in eV)	Adsorption energy E_{ads} (in K)
$N_2 \rightarrow N_{2ad}$	Fe	1.0×10^{-2}	0.176	2045
$N_{2ad} \rightarrow N_2$	Fe	$1.0 \times 10^{10} \cdot \exp(-0.3213/T_w)$	0.176	2045
$N_{2ad} \rightarrow 2N_{ad}$	Fe	$1.0 \times 10^7 \exp(-0.2902/T_w)$	2.695	31273
$H_{2ad} \rightarrow 2H_{ad}$		$0.1 \exp(-0.0311/T_w)$	0.913-1.130	10600-13110
$H_{ad} + N_{ad} \rightarrow NH_{ad}$	Fe	Not given Expression may be derived from Boudart with $\nu_D = 10^{13} s^{-1}$, $E_d = 0.2$ eV and $E_r = 0.3$ eV	1.1	12750
$H_{ad} + NH_{ad} \rightarrow NH_{2ad}$	Fe	Not given Expression may be derived from Boudart with $\nu_D = 10^{13} s^{-1}$, $E_d = 0.2$ eV and $E_r = 0.3$ eV	0.342	3970
$H_{ad} + NH_{2ad} \rightarrow NH_{3ad}$	Fe	Expression may be derived from Boudart with $\nu_D = 10^{13} s^{-1}$, $E_d = 0.2$ eV and $E_r = 0.3$ eV	0.425	4931
$NH_3 \rightarrow NH_{3ad}$	Fe	0.16	0.518	6014

One of the major drawbacks of the model proposed by Ertl and coworkers is that it does not account for a possible

Table 11 : Global model proposed by Gordiets et al.²⁴ and sticking coefficient for surface processes

	Reaction	Initial sticking coefficient on free surface	Energy(K)
S1	$N + F_v \rightarrow Nf$	Not given	Not given
S2	$H + F_v \rightarrow Hf$	Not given	Not given
S3	$N + S_v \rightarrow N(s)$	Not given	0
S4	$Nf + S_v \rightarrow N(s) + F_v$	Not given	Not given
S5	$N + N(s) \rightarrow N_2 + S_v$	6.0×10^{-5}	900
S6	$Nf + N(s) \rightarrow N_2 + F_v + S_v$	Not given	2600
S7	$H + S_v \rightarrow H(s)$	Not given	0
S8	$Hf + S_v \rightarrow H(s) + F_v$	Not given	500
S9	$H + H(s) \rightarrow H_2 + S_v$	1.2×10^{-2}	840
S10	$H + H(s) \rightarrow H_2 + F_v + S_v$	Not given	2340
S11	$N + H(s) \rightarrow NH(s)$	Not given	Not given
S12	$Nf + H(s) \rightarrow NH(s) + F_v$	Not given	4000
S13	$H + N(s) \rightarrow NH(s)$	Not given	Not given
S14	$Hf + N(s) \rightarrow NH(s) + F_v$	Not given	4000
S15	$NH + S_v \rightarrow NH(s)$	Not given	3000
S16	$NH_2 + S_v \rightarrow NH_2(s)$	Not given	3000
S17	$H + NH(s) \rightarrow NH_2(s)$	Not given	Not given
S18	$Hf + NH(s) \rightarrow NH_2(s) + F_v$	Not given	0
S19	$NH + H(s) \rightarrow NH_2(s)$	1.0×10^{-2}	3000
S20	$NH_2(s) + H \rightarrow NH_3 + S_v$	Not given	Not given
S21	$NH_2(s) + Hf \rightarrow NH_3 + S_v + F_v$	Not given	0
S22	$H + NH_2(s) \rightarrow NH_3 + S_v$	Not given	3000
S23	$H_2 + NH(s) \rightarrow NH_3 + F_s$	Not given	0

Table 12 : Simplified scheme for surface processes proposed by Carrasco et al. ^{6,105}

Reaction	Sticking Coefficient or reaction frequency (s ⁻¹)	Origin of the value
H+Fs -> H(s)	1	99
H + H(s) -> H ₂ +Fs	1.5×10^{-3}	106
N + Fs -> N(s)	1	99
N + N(s) -> N ₂ +Fs	6.0×10^{-3}	99
NH + Fs -> NH (s)	1	99
NH ₂ + Fs -> NH ₂ (s)	1	99
N + H(s) -> NH(s)	1×10^{-2}	24
H + N(s) -> NH(s)	8.0×10^{-3}	24
H + NH(s) -> NH ₂ (s)	8.0×10^{-3}	24
NH + H(s) -> NH ₂ (s)	1×10^{-2}	24
NH(s) + H(s) -> NH ₂ (s) + Fs	$10^{13} \cdot \exp(-0.5/T_w)$ T _w in eV	107, 108
H+NH ₂ (s) -> NH ₃ +Fs	8.0×10^{-3}	109
NH ₂ + H(s) -> NH ₃ +Fs	1×10^{-2}	24
H ₂ + NH(s) -> NH ₃ +Fs	8.0×10^{-4}	24
NH ₂ (s) + H(s) -> NH ₃ +2Fs	$10^{13} \cdot \exp(-0.4/T_w)$ T _w in eV	107,108

A simplified version of this model was proposed by Carrasco et al.⁶. This model listed in Table 12 also account for both LH and ER mechanisms, and essentially relies on Ertl's and Gordiet's mechanisms. The data used for surface reaction rates are based on the work of Ertl and using the surface diffusion theory described above in this section. This model was validated by comparison with experimental results in N₂/H₂ plasmas at low pressure (0.2 to 8 bar) and with low N₂ concentration. This is why we recommend it for EIRENE simulations.

V- A reduced kinetic scheme for NH₃ formation mechanism under conditions of edge plasmas

As we discussed in the previous sections of this report, taking into account all the elementary processes reported for N₂/H₂ plasmas can lead to a kinetic scheme involving more than two hundred reactions. However, since the required computation cost for models such as EIRENE may vary exponentially with the number of species and reactions, such a large kinetic model is not practical and should be therefore reduced.

Therefore, we first performed an approximate estimation of the characteristic times of the different processes. Taking into account the wide targeted ranges of plasma parameters and wall temperature, we choose to setup a kinetic scheme that ensures that each species generated in the plasma can be consumed in the gas phase. We therefore identify in a first step (step 1) the major gas phase production and consumption process for each species that were included in the model. This implies that the number of processes should be at least two times larger than the number of species, i.e., which means that we should have at least 30 processes. Of course, wall-recombination of ions, wall-de-excitation of metastables and surface chemistry for neutrals is also taken into account.

Then, in a second step (step 2) we estimated the diffusion characteristic times to the surface for the conditions of interest. We disregarded all the processes with characteristic times larger than the diffusion one at the upper limit value of the pressure. Almost all the processes involving only neutral ground state species can be disregarded on the basis of this consideration.

Then the characteristic times of the production and consumption process for each species are estimated assuming nitrogen abundance of less than 4% (2% N₂ in the plasma). We also assume that nitrogen containing species show similar densities. Hydrogen-ions, i.e., H⁺, H₂⁺ and H₃⁺, were also assumed to have similar densities. Then, in step 3, only processes with characteristic times comparable to the predominant production and consumption processes identified in step 1 are selected.

Table 13 : Example of comparative calculation of the reaction times of the NH₃⁺ production processes

Processes type	Rate constant at Te or Tgaz	Processes frequency (in s ⁻¹)	Time between each processes (in s)
H ⁺ +NH ₃ (Tgaz=300K)	3.7×10^{-09}	$3.7 \times 10^{+03}$	2.70×10^{-04}
H ₂ ⁺ + NH ₃ (Tgaz=300K)	2.0×10^{-09}	$2.0 \times 10^{+03}$	5.00×10^{-04}
e-+NH ₃ (Te=1 eV)	1.8×10^{-14}	1.8×10^{-02}	$5.43 \times 10^{+01}$
e-+NH ₃ (10 eV)	9.7×10^{-09}	$9.7 \times 10^{+03}$	1.04×10^{-04}
e-+NH ₃ (100 eV)	8.3×10^{-08}	$8.3 \times 10^{+04}$	1.21×10^{-05}
e-+NH ₃ (1000 eV)	2.4×10^{-07}	$2.4 \times 10^{+05}$	4.26×10^{-06}

We found that, very often, due to the wide range of electron temperature, at least 2-3 production processes have to be considered for each species.

As far as ionization processes are concerned, we considered classical electron impact dissociative and non-dissociative ionization of molecular hydrogen and nitrogen as well as ionization of atomic hydrogen and nitrogen. We found that NH_3^+ , NH_2^+ and NH^+ ions are mainly produced by electron-impact at high electron temperature, i.e., $T_e > 10$ eV and by charge transfer from hydrogen ions at low electron temperature, i.e., $T_e < 10$ eV. It is therefore important to consider both electron-impact and charge transfer processes in order to achieve accurate description of ammonia ionization over the very wide range of electron temperature considered in this work. We showed however that electron-impact ionization on NH_2 radical can be disregarded, NH_2^+ being mainly obtained by dissociative ionization of NH_3 .

The dissociation of ammonia into neutral fragments through electron-impact processes was found to be a possible major potential source of NH_2 and NH radical even at low electron temperature. The corresponding processes were therefore included in the proposed model. We showed however that dissociation of NH_2 and NH can be disregarded.

There is a large number of ion conversion processes. These may be of great importance in determining nitrogen-containing ion, N^+ , NH^+ , NH_2^+ , NH_3^+ , NH_4^+ and N_2H^+ populations distribution, which may in turn affect the neutral nitrogen containing species. Taking into account the large ranges of electron temperature and gas pressure, we consider that the populations of the three hydrogen ions, H^+ , H_2^+ and H_3^+ , may be significant, depending on the actual plasma conditions. We therefore take into account all the conversion processes involving hydrogen ions. These processes are very fast, i.e., their characteristic times are very small. They are likely to govern hydrogen ions population at low electron temperature and high pressure, and nitrogen ion populations at low electron temperature. Due to the low abundance of nitrogen, $\text{N}/\text{H} < 4 \times 10^{-2}$, the ion conversion process involving two nitrogen-containing ions are disregarded.

As for nitrogen metastable, we considered only $\text{N}_2(\text{A})$ and $\text{N}(\text{D})$ states. We found that they are mainly produced by electron-impact processes. They are lost by collisional quenching with H-atom, H_2 -molecule and NH_3 . Of course metastable wall-de-excitation is also taken into account.

As a result of the analysis summarized in the last paragraphs, we found that N/H plasmas chemistry may be described using 50 gas phase reactions. A further reduction to 40 reactions scheme can be performed with additional assumptions, including removal of the metastable chemistry and ions lumping. Even further reduction may be performed if information on the predominant hydrogen ion among H_3^+ , H_2^+ and H^+ is available. As a matter of fact, ion conversion processes show very similar rate constants, which prevent any possible reduction among these processes without any information on the predominance of the species involved in this process.

To conclude, we recommend a reduced mechanism involving the 50 gas phase processes, reported Table 14 to Table 20, and the 15 surface reactions proposed by Carrasco and listed in Table 12. It is possible to further reduce this mechanism to 36 gas phase reactions and 15 surface reactions mechanism by lumping the reactions in light-gray rows Table 16 and Table 17 and disregarding the dark gray reactions involving the metastable-states of the atomic and molecular nitrogen (Table 18 to Table 20).

Table 14 : Reduced kinetic scheme for ionization processes by collision with electron

Number in Appendix	Reaction	Kinetic constant in $\text{cm}^3.\text{s}^{-1}$ / Te in eV	Reference
1	$\text{N} + \text{e} \rightarrow \text{N}^+ + 2\text{e}$	$9.3 \times 10^{-9} \times \text{Te}^{0.56} \times \exp(-16.66/\text{Te})$	37
2	$\text{N}_2 + \text{e} \rightarrow \text{N}^+ + \text{N} + 2\text{e}$	$2.9 \times 10^{-9} \times \text{Te}^{0.72} \times \exp(-29.71/\text{Te})$	38
3	$\text{N}_2 + \text{e} \rightarrow \text{N}_2^+ + 2\text{e}$	$1.3 \times 10^{-8} \times \text{Te}^{0.56} \times \exp(-17.07/\text{Te})$	38
4	$\text{H} + \text{e} \rightarrow \text{H}^+ + 2\text{e}$	$1.1 \times 10^{-8} \times \text{Te}^{0.29} \times \exp(-15.28/\text{Te})$	39
5	$\text{H}_2 + \text{e} \rightarrow \text{H}^+ + \text{H} + 2\text{e}$	$9.4 \times 10^{-10} \times \text{Te}^{0.45} \times \exp(-29.94/\text{Te})$	40
6	$\text{H}_2 + \text{e} \rightarrow \text{H}_2^+ + 2\text{e}$	$2.3 \times 10^{-8} \times \text{Te}^{0.19} \times \exp(-17.87/\text{Te})$	40
7	$\text{NH} + \text{e} \rightarrow \text{NH}^+ + 2\text{e}$	$2.1 \times 10^{-8} \times \text{Te}^{0.37} \times \exp(-15.49/\text{Te})$	41
11	$\text{NH}_3 + \text{e} \rightarrow \text{NH}_3^+ + 2\text{e}$	$1.5 \times 10^{-8} \times \text{Te}^{0.4} \times \exp(-13.61/\text{Te})$	41
12	$\text{NH}_3 + \text{e} \rightarrow \text{NH}_2^+ + \text{H} + 2\text{e}$	$1.6 \times 10^{-8} \times \text{Te}^{0.34} \times \exp(-15.41/\text{Te})$	41

Table 15 : Reduced kinetic scheme for Electron-impact dissociation

Number in Appendix	Reaction	Kinetic constant in $\text{cm}^3.\text{s}^{-1}$ / Te in eV	Reference
13	$\text{H}_2 + \text{e} \rightarrow \text{H} + \text{H} + \text{e}$	$8.4 \times 10^{-8} \times \text{Te}^{-0.45} \times \exp(-11.18/\text{Te})$	40
14	$\text{N}_2 + \text{e} \rightarrow \text{N} + \text{N}(^2\text{D}) + \text{e}$	$2.4 \times 10^{-8} \times \text{Te}^{0.27} \times \exp(-15.53/\text{Te})$	38
18	$\text{NH}_3 + \text{e} \rightarrow \text{NH}_2 + \text{H} + \text{e}$	$4.2 \times 10^{-8} \times \text{Te}^{-0.19} \times \exp(-7.59/\text{Te})$	44
ZE	$\text{NH}_3 + \text{e} \rightarrow \text{NH} + 2\text{H} + \text{e}$	$1.3 \times 10^{-8} \times \text{Te}^{0.38} \times \exp(-11.06/\text{Te})$	

Table 16 : Reduced kinetic scheme for Dissociative recombination processes*

Number in Appendix	Reaction	Kinetic constant in $\text{cm}^3.\text{s}^{-1}$ / Te in eV	Reference
20	$\text{H}_2^+ + \text{e} \rightarrow \text{H} + \text{H}$	$1.60 \times 10^{-8} \times (\text{Te}/0.026)^{-0.43}$	47
21	$\text{H}_3^+ + \text{e} \rightarrow 3\text{H}$	$4.36 \times 10^{-8} \times (\text{Te}/0.026)^{-0.52}$	48
22	$\text{H}_3^+ + \text{e} \rightarrow \text{H}_2 + \text{H}$	$2.34 \times 10^{-8} \times (\text{Te}/0.026)^{-0.52}$	45
23	$\text{N}_2^+ + \text{e} \rightarrow \text{N}(^2\text{D}) + \text{N}$	$1.70 \times 10^{-7} \times (\text{Te}/0.026)^{-0.3}$	49
24	$\text{NH}^+ + \text{e} \rightarrow \text{N} + \text{H}$	$4.30 \times 10^{-8} \times (\text{Te}/0.026)^{-0.5}$	47
25	$\text{NH}_2^+ + \text{e} \rightarrow \text{NH} + \text{H}$	$1.02 \times 10^{-7} \times (\text{Te}/0.026)^{-0.4}$	50
27	$\text{NH}_3^+ + \text{e} \rightarrow \text{NH} + 2\text{H}$	$1.55 \times 10^{-7} \times (\text{Te}/0.026)^{-0.5}$	47
28	$\text{NH}_3^+ + \text{e} \rightarrow \text{NH}_2 + \text{H}$	$1.55 \times 10^{-7} \times (\text{Te}/0.026)^{-0.5}$	47
29	$\text{NH}_4^+ + \text{e} \rightarrow \text{NH}_3 + \text{H}$	$8.49 \times 10^{-7} \times (\text{Te}/0.026)^{-0.6}$	50
31	$\text{N}_2\text{H}^+ + \text{e} \rightarrow \text{N}_2 + \text{H}$	$5.13 \times 10^{-8} \times (\text{Te}/0.026)^{-0.72}$	52

* Processes that may be lumped : 21 with 22, and 27 with 28.

Table 17 : Reduced kinetic scheme for ion/neutral collisional processes

Number in Appendix	Reaction	Kinetic constant in $\text{cm}^3.\text{s}^{-1}$ / Tg in eV	Reference
32	$H^+ + NH_3 \rightarrow NH_3^+ + H$	$3.7 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	58
ZL	$H^+ + NH_2 \rightarrow NH_2^+ + H$	$2.9 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
ZM	$H^+ + NH \rightarrow NH^+ + H$	$2.1 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	
33	$H_2^+ + H \rightarrow H_2 + H^+$	6.4×10^{-10}	60
34	$H_2^+ + H_2 \rightarrow H_3^+ + H$	2.00×10^{-09}	53
35	$H_2^+ + NH_3 \rightarrow NH_3^+ + H_2$	5.70×10^{-09}	61
36	$H_2^+ + N_2 \rightarrow N_2H^+ + H$	2.00×10^{-09}	53
ZN	$H_2^+ + N \rightarrow N^+ + H_2$	5.00×10^{-10}	62
ZO	$H_2^+ + N \rightarrow NH^+ + H$	1.9×10^{-09}	59
ZR	$H_2^+ + NH_2 \rightarrow NH_2^+ + H_2$	$2.1 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
ZT	$H_2^+ + NH \rightarrow NH^+ + H_2$	$7.6 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59
39	$H_3^+ + NH_3 \rightarrow NH_4^+ + H_2$	$4.40 \times 10^{-09} \times (Tg/300)^{-0.5}$	53
ZU	$H_3^+ + NH_2 \rightarrow NH_3^+ + H_2$	$1.8 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
ZV	$H_3^+ + NH \rightarrow NH_2^+ + H_2$	$1.3 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
40	$H_3^+ + N_2 \rightarrow N_2H^+ + H_2$	1.90×10^{-09}	53
41	$N^+ + H_2 \rightarrow NH^+ + H$	5.00×10^{-10}	53
ZZ	$N^+ + H \rightarrow H^+ + N$	2.00×10^{-10}	62
46	$NH^+ + H_2 \rightarrow NH_2^+ + H$	1.00×10^{-09}	53
50	$NH_2^+ + H_2 \rightarrow NH_3^+ + H$	2.00×10^{-10}	53

* Processes that may be lumped: 32 with 35 ; ZL with ZR and ZU ; ZM with ZT and ZV.

Table 18 : Electronic collisional processes and rate coefficients for the formation of N and N₂ metastable states

Number in Appendix	Reactions	Kinetic constant in cm ³ .s ⁻¹ / Te in eV	Ref.
EK	$N_2 + e \rightarrow N_2(A) + e$	$1.4 \times 10^{-08} \times \exp(-6.325 \times 10^{-04}/Te)$	35
EM	$N + e \rightarrow N(^2D) + e$	$1.0 \times 10^{-8} \times Te^{-0.36} \times \exp(-7.15 \times 10^{-05}/Te)$	35

Table 19 : Processes and rate coefficients for the quenching and the excitation of N₂ electronic levels by heavy particle collisions

Reaction of main species with N ₂ metastables			
Number in Appendix	Reactions	Kinetic constant in cm ³ .s ⁻¹ / Tg in eV	Ref.
DB	$N_2(A) + N(^4S) \rightarrow N_2(X) + N(^2P)$	$1.80 \times 10^{-09} \times (Tg/0.026)^{-0.67}$	68-70
DD	$N_2(A) + NH_3 \rightarrow N_2(X) + H + NH_2$	8.51×10^{-11}	72
DE	$N_2(A) + H \rightarrow N_2(X) + H$	2.09×10^{-10}	73
DF	$N_2(A) + H_2 \rightarrow N_2(X) + 2H$	$2.00 \times 10^{-10} \times \exp(-0.3016/Tg)$	74,75

Table 20 :Rate coefficients for the quenching of metastable nitrogen atoms by collisions with atoms and molecules.

Reaction of main species with N metastables			
Number in Appendix	Reactions	Kinetic constant in cm ³ .s ⁻¹ / Tg in eV	Ref.
EA	$N(^2D) + H_2 \rightarrow NH + H$	2.29×10^{-12}	84
EC	$N(^2D) + NH_3 \rightarrow NH + NH_2$	1.10×10^{-10}	83

Important note: The figures showing the evolution of the collision frequency as a function of Te, Tg or Tw of all the processes described in this report were plotted in the appendix. All reactions are identified by a reference number reported in the column named "Number in appendix" for each table. For most of the reactions, 2 or 3 data sources are compared

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